Language
Understanding
$\lambda(\mu)$ Laboratory

String Similarity Measures for Myanmar Language

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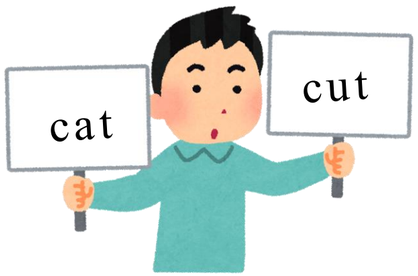
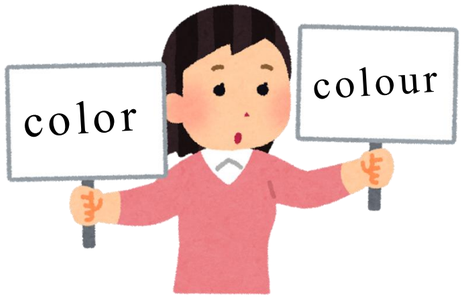
APRIL 4, 2026

Background

- String similarity measures **how similar** the two strings are.



- In language processing, it helps with searching words, correcting typos, and understanding similar meanings.



- In general, there are **three** main types of metrics to measure string similarity,
  - Edit-distance based metrics
  - Token-based metrics
  - Sequence-based metrics

Six similarity measures

- Edit-distance based metrics
  - Levenshtein Distance
  - Damerau-Levenshtein Distance
  - Hamming Distance

- Token-based metrics
  - Jaccard Similarity
  - Cosine Similarity

- Sequence-based metrics
  - JaroWinkler Similarity

Levenshtein Distance

- measure similarity by counting operations needed to transform one string into another
- Given two character strings s1 and s2, the edit distance between them is the minimum number of edit operations required to transform s1 into s2.
- The edit operations are
  - Insert
  - Delete
  - Substitute

S1 = hot
S2 = pot

h o t
substitute ↓
p o t

Edit distance = 1

S1 = kitten
S2 = sitting

k i t t e n
substitute ↓ substitute ↓ insert ↓
s i t t i n g

Edit distance = 3

Damerau-Levenshtein Distance

- differs from the Levenshtein distance by including transpositions among its allowable operations in addition to three classical single character edit operations

- The edit operations are
  - Insert
  - Delete
  - Substitute
  - Transpose (swap adjacent characters)

S1 = form
S2 = from

f o r m

transpose

f r o m

Levenshtein distance = 2

Damerau-Levenshtein distance = 1

Hamming Distance

- The Hamming distance between two strings of equal length measures the number of positions with mismatching characters.

S1 = hot
S2 = pot

	h	o	t
different	↓	same	
	p	o	t

Hamming distance = 1

Jaccard Similarity

- measures similarities between sets

$$\text{Jaccard Similarity} = \frac{|A \cap B|}{|A \cup B|}$$

Sentence 1: I love data science
Sentence 2: I love machine learning

$\xrightarrow{\text{tokenize}}$

A = {I, love, data, science}
B = {I, love, machine, learning}

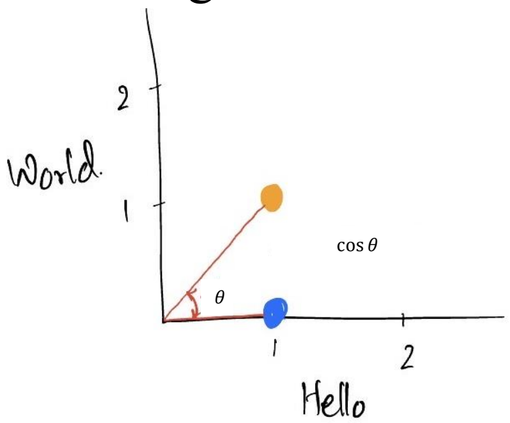
$A \cap B = 2$
$A \cup B = 6$

Jaccard Similarity = 2/6 = 0.33

Cosine Similarity

- how similar two texts are by looking at the angle between their word or character vectors.

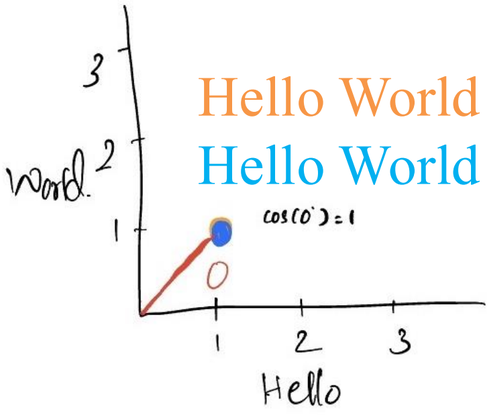
$$\text{Cosine Similarity} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$



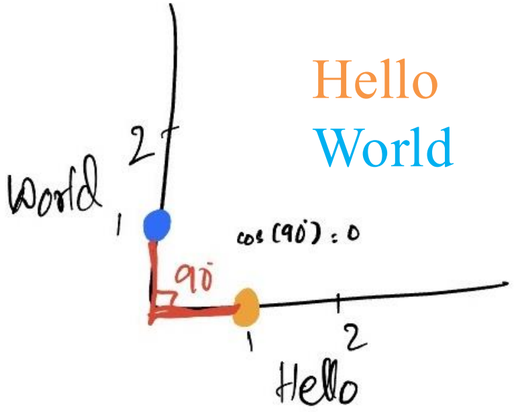
Hello World
Hello

Hello	World
1	1
1	0

$$\text{Cosine Similarity} = \frac{(1 \times 1) + (1 \times 0)}{\sqrt{1^2 + 1^2}\sqrt{1^2 + 0^2}} = 0.71$$



Hello World
Hello World



Hello
World

Jaro Winkler Similarity

- string metric measuring an edit distance between two sequences.
- The value ranges from 0 to 1. (0 = completely different, 1 = exactly the same)

$$\text{Jaro}(s1, s2) = \frac{1}{3}\left(\frac{m}{|s1|} + \frac{m}{|s2|} + \frac{m-t}{m}\right)$$

- m = number of matching characters
- t = number of transpositions
- |s1|, |s2| = lengths of the two strings

s1 = ABC
s2 = CBA

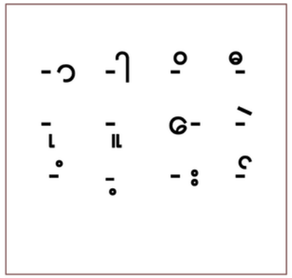
m = 1, t = 0, s1 = s2 = 3

$$\text{Jaro (s1,s2)} = \frac{1}{3}\left(\frac{1}{3} + \frac{1}{3} + \frac{1-0}{1}\right) \approx 0.56$$

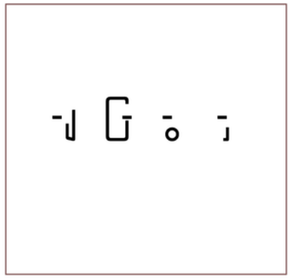
Contribution

- Myanmar Language
  - is a tonal language
  - not only consonant can form a word
    - “က” - “dance” in English
  - but also one syllable can form a word
    - “ကြိုက်” - “like” in English
  - many phonetically similar sounds of characters and words
- As existing string similarity methods were designed for English or non-tonal languages, so they **fail** to handle Myanmar language properly.

Basic Consonant



Vowels



Medials

ကျောင်း

Example with combination of consonants, vowels and medials

Propose

three mapping approaches with the existing **string similarity metrics** for Myanmar Language

Proposed Mapping

- Phonetic Mapping
- Sound Mapping
- Vowel Position Mapping

Six similarity measures

- Edit-distance based metrics
  - Levenshtein Distance
  - Damerau-Levenshtein Distance
  - Hamming Distance
- Token-based metrics
  - Jaccard Similarity
  - Cosine Similarity
- Sequence-based Methods
  - JaroWinkler Similarity

Soundex

- Soundex algorithm is a phonetic algorithm.
- It is based on how close two words are depending on pronunciation.
- It converts words into codes based on how they sound, not how they are spelled.

Flower = F406

Flour = F406

Letters	Code
A,E,H,I,O,U,W,Y	0
B,F,P,V	1
C,G,J,K,Q,S,X,Z	2
M,N	5
R	6

Based on the idea of Soundex algorithm, there are three proposed mappings for Myanmar Language.

Proposed Mapping - Phonetic Mapping

Letters	Code	Letters	Code
ကခ	က	့ ့	(delete)
ဂဃ	ဂ	ဣ ဤ ၏ ဦ ဉ်	i
စဆ	စ	ဂ် ပ် တ်	d
ဇ ဈ	ဇ	နှ် မ် ံ	n
ဋ ဌ	တ	ဲ ရ်	e
ဍ ဎ	ထ	ဥ ဦ ု ူ	u
ဏ ဝ	ဒ	တ ါ	r
ဏ	န	ဧ ေ	a
ဒ ဓ	ဒ	့ း	(delete)
ပ ဖ	ပ	္	(delete)
ဗ ဘ	ဘ	သြ ော် သြ ော်	o
ယ ရ	ရ	ဍ ဏ ဋ	ဍ
လ ဠ	လ	၊ ။	s
သ ဿ	သ	ိ ် င ဉ်	in
ျ ြ	y	?!.*-=\#"<>{}[],+-	s

Table : Phonetic Mapping

- Same pronunciation words are grouped together.
- For example, ကလေး and ခလေး have same pronunciation.
- Some diacritics such as ့ (WaHswe) and ့ (Ha Hto), tone marks such as ့ (Aukmyit), ္ (Myanmar sign Virama) are considered to be removed.
- Source String = ပစ်စီး
- Target String = ပစ္စည်း
- Encoded Source String using Phonetic Mapping = ပစစi
- Encoded Target String using Phonetic Mapping = ပစစi

Proposed Mapping - Phonetic Mapping

Source String Target String	ပစ်စီး ပစ္စည်း	ပစစi ပစစi
Levenshtein Distance	3	0
Damerau-Levenshtein Distance	3	0
Hamming Distance	4	0
Jaro Winkler Distance	0.7968	1
Cosine Similarity	0.4999	1
Jaccard Similarity	0.4444	0.6

Table : Calculation of string similarity between two strings with Phonetic Mapping

Proposed Mapping - Sound Mapping

Letters	Code	Letters	Code
ကခဂဃငဟအ	က	့ ္	(delete)
ညဉ	ည	ဣ ဤ ၏ ဦ ဧ ည်	i
စဆဇဈ	စ	ဋ် ပ် တ်	d
ဌဍဎဏ ၊ ၎ ၊ ၎ ၊ ၎

Proposed Mapping - Sound Mapping

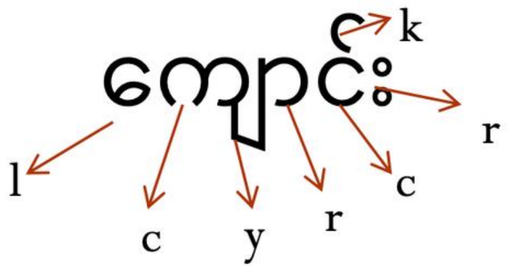
Source String Target String	ဖုရား ဘုရား	ပရr ပရr
Levenshtein Distance	1	0
Damerau-Levenshtein Distance	1	0
Hamming Distance	1	0
Jaro Winkler Distance	0.8667	1
Cosine Similarity	0.4999	1
Jaccard Similarity	0.6667	1

Table : Calculation of string similarity between two strings with Sound Mapping

Proposed Mapping - Vowel Position Mapping

Letters	Code	Letters	Code
a-z A-Z	F	ကာ-အ	c
ျ ြ	y	့	p
ေ	l	ာ ါ ိ ီ ု ူ ိ	r
ိ ီ ဲ ံ	u	့ ့ ့ ့ ့ ့ ့ ့	d
်	k	္	s
ဣ ဤ ဥ ဦ ဧ ြ ြ ော် ော ် ် ် ်	i	?!.*-=\#"<>{}[],+-	$
၀-၉	n	0-9	D

Table : Vowel Position Mapping



- The third mapping based on the syllable formation of Myanmar language.
- The vowels written on the left side of the consonant are under the left (l) group, the right side vowels are under the (r) group, the upper vowels are under the (u) group, the lower vowels are under the (d) group respectively.

Proposed Mapping - Vowel Position Mapping

Source String Target String	ကျောင်း ချောင်း	cylrckr cylrckr
Levenshtein Distance	1	0
Damerau-Levenshtein Distance	1	0
Hamming Distance	1	0
Jaro Winkler Distance	0.9047	1
Cosine Similarity	0.4999	1
Jaccard Similarity	0.75	1

Table : Calculation of string similarity between two strings with Vowel Position Mapping

Datasets and Evaluation

Spelling Mistake Confusion Pair

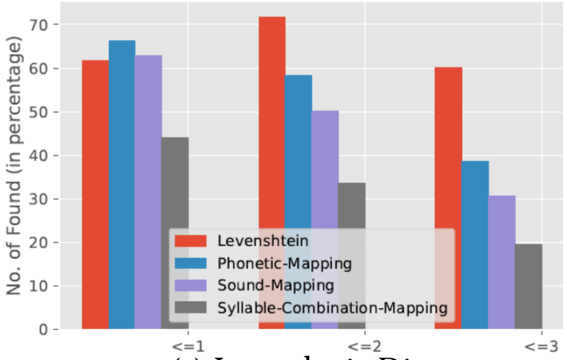
- Develop based on the real world spelling errors
- Collect from Myanmar web news and social media website (BBC,VOA, Facebook)
- 2,381 pairs in total (i.e. 4,762 words)

Word Similarity Data Set

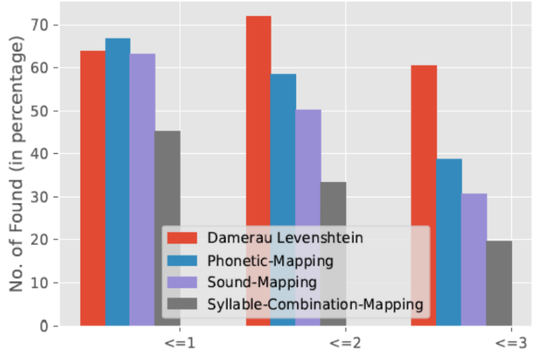
- Develop similar pronunciation data set to measure how well the
- similarity scores provided by three mappings
- Manually added the correct words together with homophone
- and rhyme words
- 200 pairs in total (i.e. 1,000 words)

Datasets and Evaluation

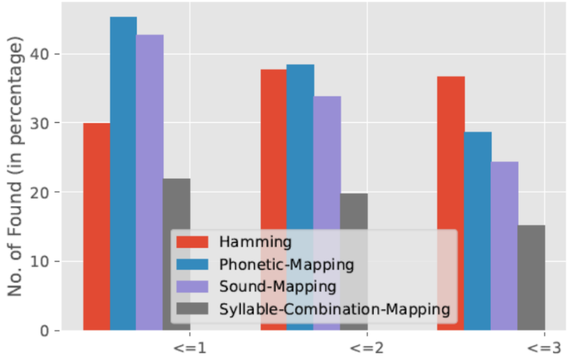
- For the evaluation, string similarity on each pair of data set is measured.
- After that, the original data is encoded or converted with three proposed mappings and string similarity is measured again.
- Finally, count the correct words or similar words based on the three thresholds <=1,<=2 and <=3 for “Levenshtein”, “Damerau Levenshtein” and “Hamming Distance” measures and >=0.9,>=0.7 and >=0.5 “Jaro Winkler”, “Cosine” and “Jaccard” distance measures.



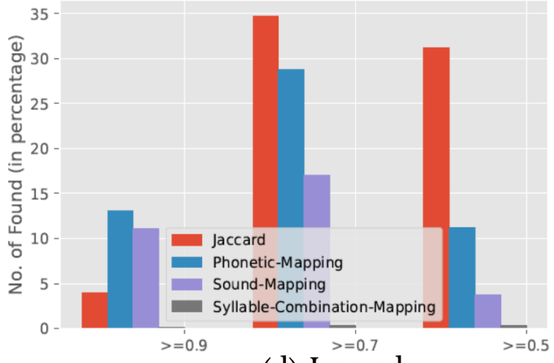
(a) Levenshtein Distance



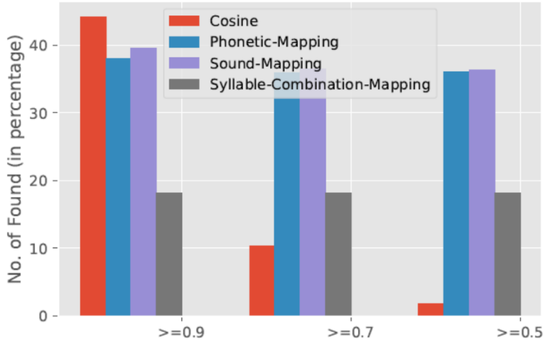
(b) Damerau-Levenshtein Distance



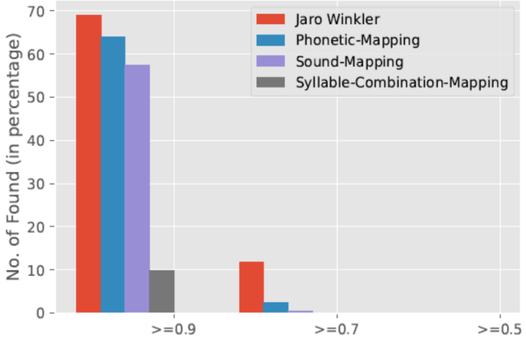
(c) Hamming Distance



(d) Jaccard



(e) Cosine Similarity

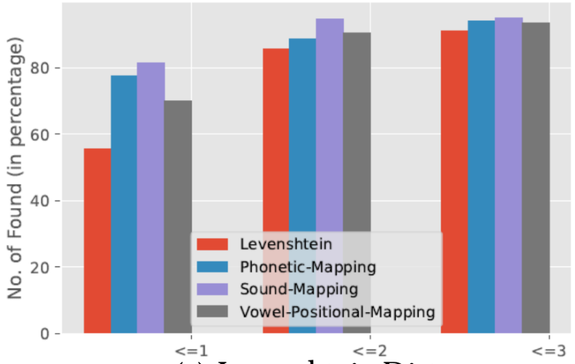


(f) Jaro Winkler

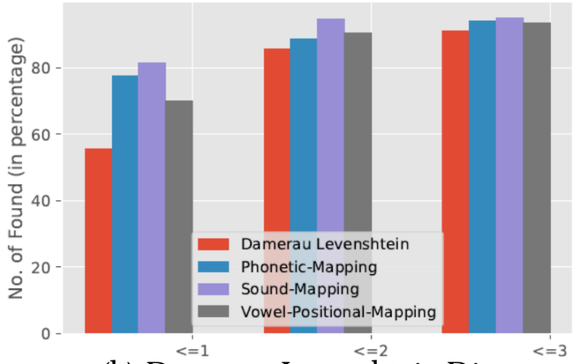
Figure : Results with the spelling-mistake confusion dataset

- According to the results, Phonetic and Sound mappings are applicable for string similarity measurement on spelling mistake confusion words.

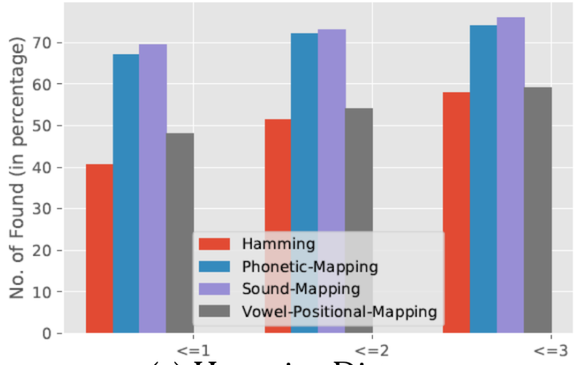
- It is assumed to be very useful for evaluating on three mappings but this data has few homophone and rhyme words, so it is not suitable for measuring pronunciation similarity.



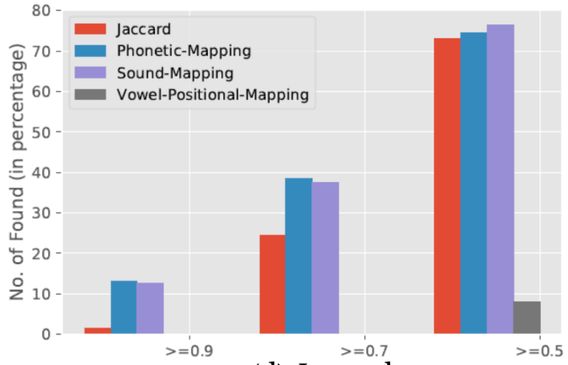
(a) Levenshtein Distance



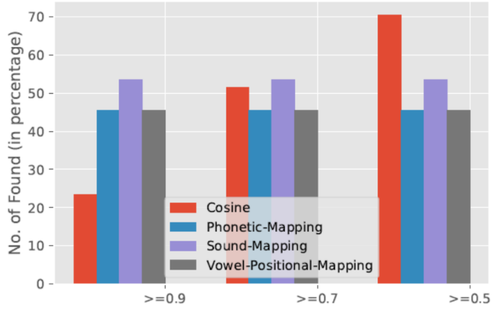
(b) Damerau-Levenshtein Distance



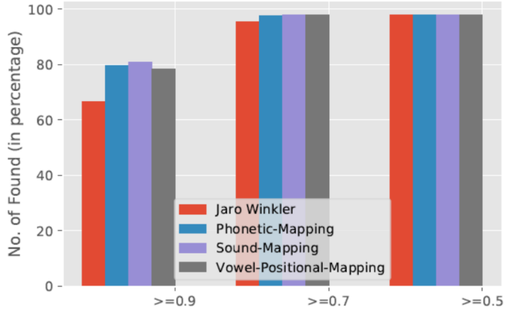
(c) Hamming Distance



(d) Jaccard



(e) Cosine



(f) Jaro Winkler

Figure : Results of the Similar Pronunciation Dataset

- As expected, our proposed two mappings: Phonetic Mapping and Sound Mapping achieved highest number of founds for all thresholds of Levenshtein, Damerau Levenshtein, Hamming, Jaro Winkler, Cosine and Jaccard distance measures.

- Additionally, the Vowel Positional Mapping also gives the highest results for existing five distance measures except for the Jaccard distance measure.